Unemployment and Crime in Turbulent times - Panel Evidence from the US State Data 2006-2016 and the California County Data 2012-2017

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Abstract

This applied paper attempts to investigate whether a potential relationship exists between unemployment and violent/property crime, and if so then to what extent? The analysis begins with the hypothesis that a positive one exists and uses a fixed effects technique to empirically test it. Separate panel regressions are run for both violent and property crime for all the US states[1] during the 2006-2016 period and the counties in California in the 2012-2017 time frame. These are a total of 4 regressions and results from 3 of them reject the hypothesis of a positive relationship. Using a 95% confidence level throughout, this paper finds;

1. For states the relationship is negative with both violent and property crime.
2. For counties the relationship is negative with violent crime and positive with property

Introduction

The motivation behind this paper is to study the crime-unemployment relationship, with an expectation that there is a positive relationship between the two. To empirically test this hypothesis, a fixed effects technique is used to evaluate the impact of unemployment on property and violent crime, for both states and the California counties. A total of 4 panel regressions are run, one for each type of crime (violent/property) at each level (state/county). An extensive list of control variables, including economic conditions and demographics are used. After detailed data analysis, this paper chooses the 2006-2016 time frame for the state-level regressions and the state of California along with the 2012-2017 period for the county-level ones.

Throughout this paper, a 95% confidence interval is used to determine statistical significance. At this confidence interval, this paper identifies a significant relationship between unemployment and crime throughout. For states, a 1 unit increase in the unemployment rate is associated with a 5.6 unit decrease in the violent crime rate and a 16.9 unit decrease in the property crime one. Both the state crime rates are per 100,000 residents. For the county level analysis, a 1 unit increase in the unemployment rate is associated with a 9.2 unit decrease in the violent crime rate, and a 60.2 unit increase in the property crime one. Both the county crime rates are per 1000 residents.

Hence, for 3 of the 4 regressions (both the state ones and the county violent crime one), the results are negative, thus rejecting the initial hypothesis of a potential positive relationship. Only the result from the county property crime regression is positive and so supports the hypothesis.

[1] Including the District of Columbia and Puerto Rico

The analysis in this paper, albeit comprehensive, suffers from the limitations of data unavailability, non-linearity and lack of theoretical contexts. Further review is recommended to cater to these short-comings.

The paper proceeds as follows:

- The first section deals with data. The paper first discusses the data for the state-level analysis, and then proceeds to discuss the county level analysis.
- Then the paper proceeds to discuss the model and regression results. The paper first discusses the results for the state regressions and the proceeds to the county ones.
- In the end, the paper concludes with limitations and recommendations for further review.

Data

The independent variables in this analysis are the violent and property crime rates. This paper defines violent crime as a crime in which an offender or perpetrator employs (or threatens to) the use of harmful force upon the victim[2] (e.g. murder, assault, rape, and assassination). This paper defines property crime as a crime involving private property. This includes crimes such burglary, larceny, theft, motor vehicle theft, arson, shoplifting, and vandalism.

The paper analyzes crime on two different levels; at the state level and the county level. Data for the state level crime rates (both violent and property) was obtained from the FBI- Uniform Crime Reporting Program. The state crime rates are crime cases per 100,000 residents. Data for the county level crime rates was sourced from the California Department of Justice. The county crime rates are crime cases per 1000 residents.

The dependent variable in this analysis is the Unemployment rate. Data for the state level unemployment rate was sourced from the Bureau of Labor Statistics. Data for the county level unemployment rate was obtained from the Employment Development Department of the State of California.

The analysis done in this paper is an empirical one and all of the variables used are quantitative and continuous. All variables are yearly in frequency and all of them are expressed in terms of population (percentages/rates or per capita). Expressing in terms of population, offers the advantage of converting all discrete variables to continuous, thereby allowing the use of a regression analysis instead of a classification one. For the state-level analysis missing data was taken care of by filling in with average values. For the county-level analysis missing data was accounted for by dropping values (the only missing data was for the Alpine county which was wholly dropped).

This paper first discusses data for the state level analysis, and then for further investigation on a more granular basis, it shifts gears towards the California counties.

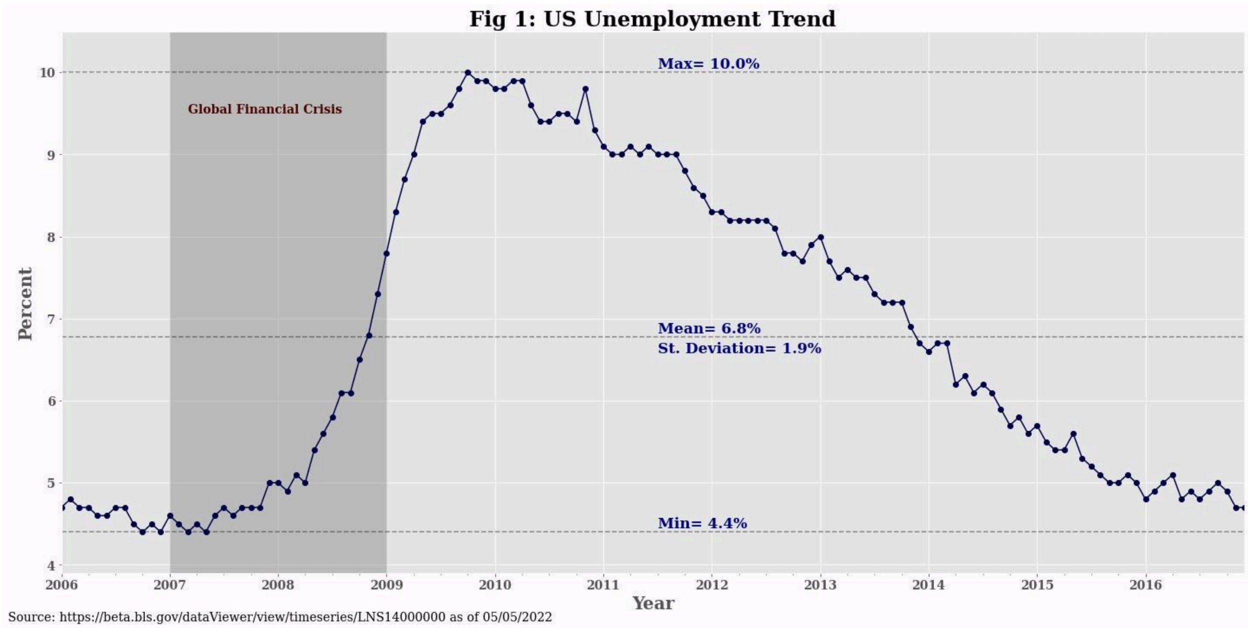
[2] "CRIME IS A RESULT OF SOCIAL AND ENVIRONMENTAL FACTORS AND NOT ON BIOLOGICAL FACTORS". ResearchGate. Retrieved 2022-02-17.

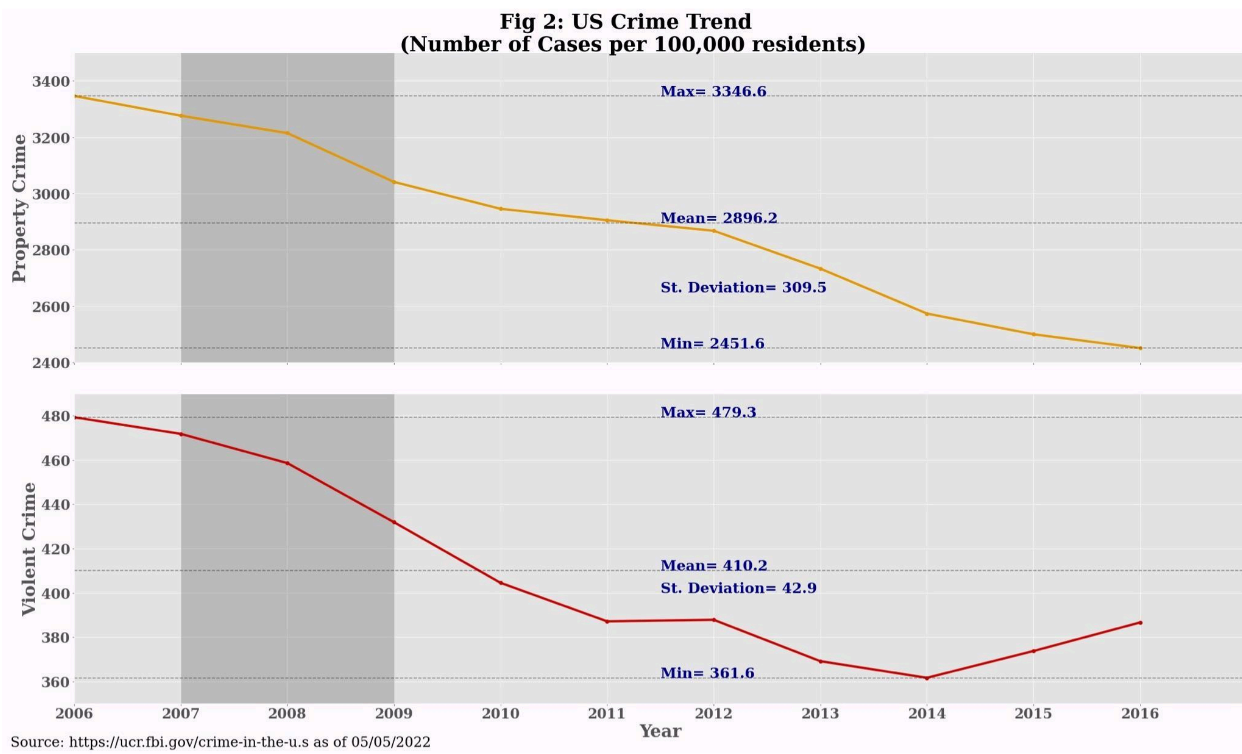
State Level Analysis

The state-level analysis looks at data for all of the US States (including the District of Columbia and Puerto Rico) for the years 2006-2016. This time period was chosen after a careful analysis. The decision to start with 2006 was made to include the Global Financial Crisis of 2007-2008. Since the onset of this crisis, a large number of workers have lost their jobs and until the end of 2016 the unemployment rate has mostly remained above the 4.4% value observed at the end of 2006 (See Figure 1).

Additionally the European crisis, in late 2009 also had rippled effects on the U.S economy, further making this period interesting to analyze. Initially, the authors aimed to extend the time period till 2020, however due to lack of official data from the FBI the analysis only goes as far as 2016.

Figure 1 below shows that from the end of 2008 to the end of 2014 the unemployment rate has remained higher than the overall average value of 6.8%. It attained an all time high of 10% in the last quarter of 2009. During the 2006-2016 period, unemployment has greatly varied and both positive (2007-late 2009) and negative (2010-2016) trends are observed. Crime rates, however, have mostly been following a general downward trend (See Figure 2). Post 2010- all three rates are moving down in the same direction. Hence it is worth studying whether empirical evidence can be found to prove a positive relationship between unemployment and the crime rates.





This paper includes an extensive set of control variables, which include both economic conditions and demographics. Multiple candidates were considered and only some of them were retained. Both the variables retained and omitted from the final analysis are discussed below.

Control variables retained

Population density: This paper defines population density as the ratio of population and land area (km^2). Crime rate in more densely populated areas tends to be higher in general and since geographical areas differ vastly in terms of population density the paper includes this as a control variable. Data for the population density variable has been sourced from the United States Census Bureau.

Poverty rate: The poverty rate variable is the ratio of individuals below the poverty level and the total population in a state. By including this variable, this paper aims to test whether the lack of sufficient income or material possessions for needs, is correlated with crime incidence.

Data for the poverty rate variable has been obtained from the United States Census Bureau. The source uses a poverty index devised by a Federal Interagency Committee to set "poverty thresholds" in order to determine whether families and unrelated (single) individuals are classified as being above or below the poverty level. The thresholds are set taking into account family sizes, number of children, and ages of the family members (or single individual)[3]. These thresholds are updated on an annual basis to reflect changes in the Consumer Price Index. For context, the 2016 poverty threshold for a family of four with two children under 18 was $24,339.

Full-time state law enforcement employee rate: This paper defines this variable as the total number of law enforcement employees per capita in a state. Whether the presence of such employees acts as a deterring factor for crime or not has often been a point of contention. This paper aims to include this variable as a control variable in order to test its significance in the context of violent and property crime.

Data for the full-time law enforcement employee rate has been sourced from the FBI- Uniform Crime Reporting Program. The sources defines law enforcement employees as employees who are ordinarily accompanied by a firearm and badge, have full powers to arrest and are funded by government funds reserved for sworn law enforcement representatives[4].

Per capita income: As shown above in Figure 2, overall crime in the US appears to fall during the recessionary period of the Global Financial Crisis. This paper aims to empirically evaluate this by including per capita income as a control variable. This paper obtains the data for per capita income from the Federal Reserve Bank of St. Louis.

Gender : This paper analyzes two demographic variables. These are the percentage of males (as a percentage of total population) in the 15-25 age group and percentage of males in the 25-39 age group. This paper looks at males within certain age brackets as crime statistics have shown that within the United States males compose of a majority of persons arrested (Figure 3). We also see that for both violent and property crime around 70% of arrests are either from the 15-24 or the 25-39 year age group (See Figure 4).

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[3] U.S. Census Bureau. (2021). *Current Population Survey: 2020 Annual Social and Economic (ASEC) Supplement*. https://www2.census.gov/programs- surveys/cps/techdocs/cpsmar20.pdf

[4] FBI. (2017, May 15). *Police employee data*. FBI. Retrieved May 7, 2022, from https://ucr.fbi.gov/crime-in-the-u.s/2016/crime-in-the-u.s.-2016/topic-pages/police-employees

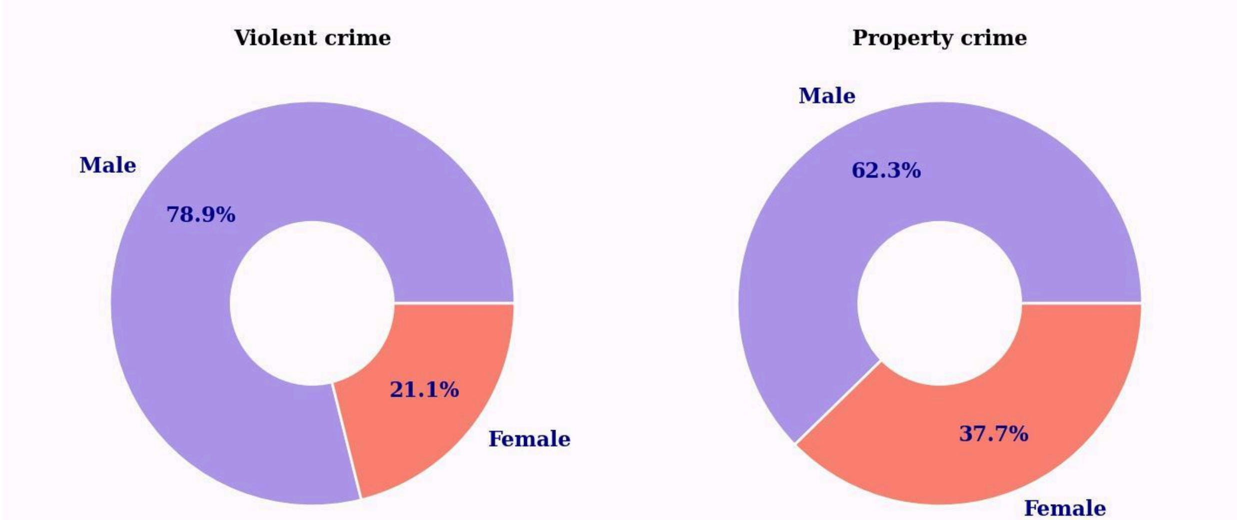


Figure 3: Total Arrests by Gender in 2019

Source: https://ucr.fbi.gov/crime-in-the-u.s
as of 04/19/2022

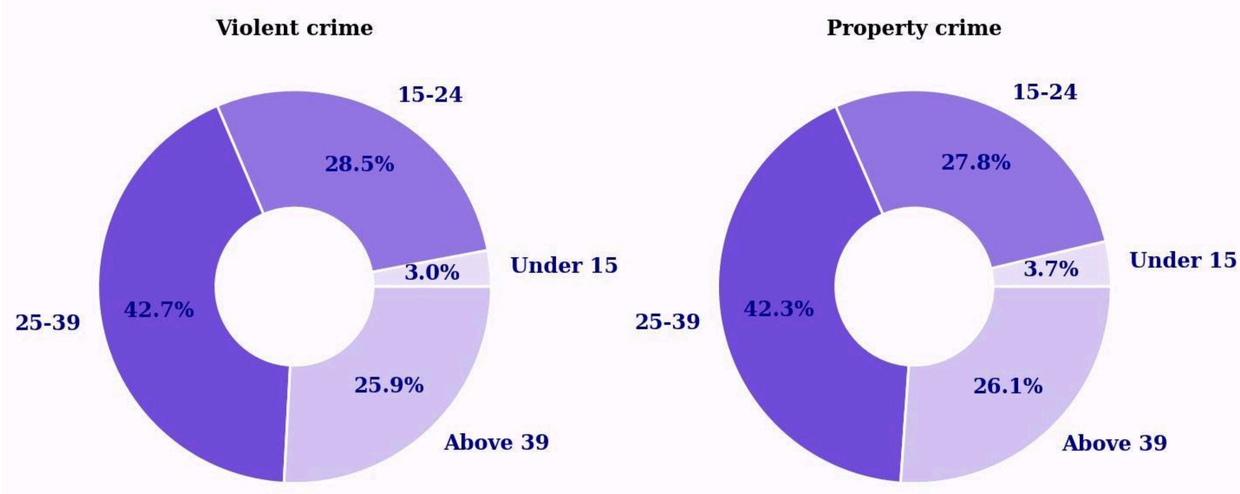


Figure 4: Total Male Arrests by Age in 2019

Source: https://ucr.fbi.gov/crime-in-the-u.s
as of 04/19/2022

By including these two age groups in the regression analysis, this paper looks at their association with crime rates. The data for these demographic variables has been sourced from the United States Census Bureau. The sources provide data as both 1 year and 5 year estimates. 1 year estimates are based on data collected over 12 months and 5 year estimates are based on 60 months. Since, 1 year estimates provide more granularity, therefore the paper uses 1 year estimates for the state analysis.

Census data provides total population in a particular area, and within that total males and females. It then divides total individuals in each gender category into consecutive 5 year age brackets (starting from under 5 years and ending with 85 years and above). Data from multiple age brackets is combined to calculate the total number of males within our age brackets of interest. These numbers are then divided by total population to calculate the rates used in the regressions.

Control variables omitted

Multiple variables were initially considered as potential candidates for the final list of controls. However, as the analysis progressed on, they were omitted/excluded from being used. These, along with the reasons to exclude them, are discussed below.

Median Household Income by State: This variable has commonly been used to represent the general economic status of a household in a state or to track economic trends. However, this variable was excluded as it did not appear to have a direct and relevant impact on crime.

State minimum wage: This variable was initially considered to be included as prior literature often focused on its relationship with crime rates. However, this paper excludes this from the analysis as the authors were not able to find complete and consistent data from all states across all the years. Data was inconsistent because some states (e.g New York) had higher minimum wages than the federal minimum wage, some (e.g Texas) had the same, some (e.g Wyoming) had lower ones (and so federal laws applied there) and some (e.g Mississippi) did not have any minimum wage (again the federal law applied).

Children in poverty (100 percent poverty) in the United States: As shown in Figure 4 above, children make up a very small proportion of individuals arrested for committing property and violent crime. Therefore, this variable was not considered anymore.

Average annual precipitation by state: Evaluating the impact of precipitation or other climate related factors such as ground temperature and extreme weather events would be an interesting topic to consider. However, due to the lack of easily accessible quantified data this variable could not be included in the analysis.

Race: A lot of prior literature looked at how race intersects with crime. This paper excluded this variable as it was suspected to result in spurious correlation. There is not enough substantial evidence to prove that race intersects with crime. Correlation between the two is likely the result of other socio-economic factors such as poverty and income which have already been included in the analysis.

Descriptive statistics for the state level dependent, independent and control variables retained are shown below:

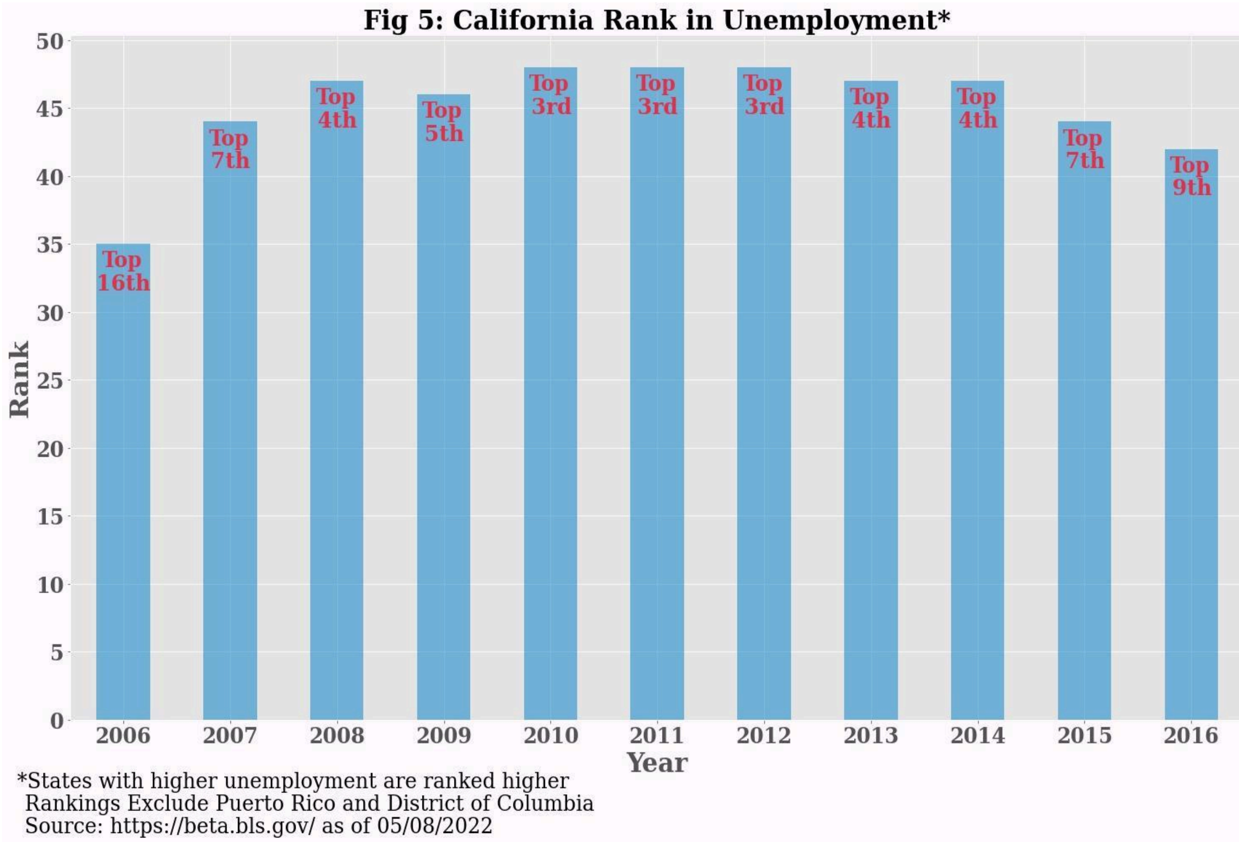
Table 1: State Descriptive Statistics

Variable Name	Mean	Standard Deviation	Minimum	Maximum	Median
Violent crime rate	389.3	198.17	99.3	1508.4	343
Property crime rate	2843.12	744.58	1031.9	5182.5	2766
Unemployment	6.48	2.43	2.6	16.4	6
Population density	134.09	483.55	0.39	3848.42	35
Poverty rate	13.28	3.38	5.4	23.1	13
Law enforcement employee rate	4.64E-04	5.39E-04	8.30E-07	6.19E-03	3.59E-04
Per Capita income	42875.9	8001.89	27981	76623	41324
% Males 15-24 years	7.18	0.4	6.06	8.83	7
% Males 25-39 years	9.81	0.91	8.13	14.92	10

County Level Analysis

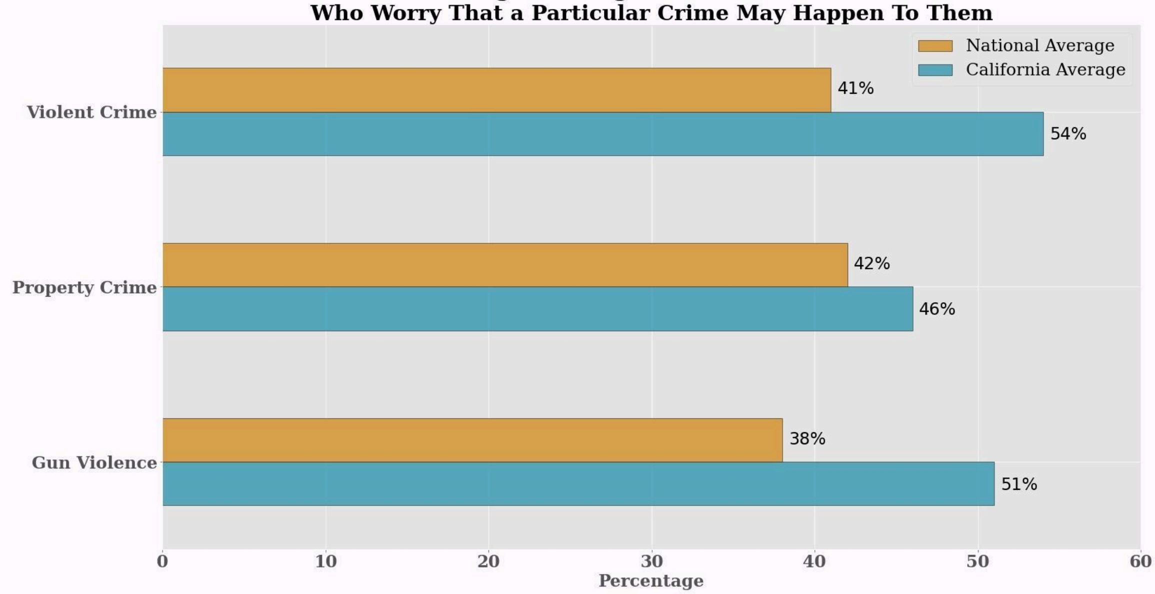
As an additional, more granular, step this paper looks at the relationship between property/violent crime and unemployment at the county-level for California. There were multiple reasons to do this. Firstly not many studies have done so at such a granular level and so this paper aims to fill in the additional gap. Secondly, counties (within a particular state) are likely to be more homogeneous than states and therefore a county-level analysis offers more depth and nuance.

The state of California was chosen for multiple reasons. California's population is the largest amongst all U.S states and is also very racially diverse, thus being an interesting candidate to analyze. Additionally, unemployment in the state of California has quite frequently been much higher than other states. Figure 5 below, shows how California ranked in unemployment in each year from 2006 to 2016. In this figure, states with higher unemployment rates are ranked higher. California has consecutively been ranked in the top 5 states from 2008-2014 (7 out of 11 years in the state sample). It has had the third highest unemployment rate (with a rank of 48) from 2010-2012. In all 11 years, California has had a rank higher than the midpoint value of 25.



Furthermore, California has had relatively high levels of concern with respect to violent crime, property crime and gun violence. Figure 6, uses data from the State of Safety in America (June 2021) report by SafeWise to compare the California average with the National average for the average percentage of residents who worry that a particular crime may happen to them. The figure shows that 54% of Californians are concerned about violent crime, 46% about property crime and 51% about gun violence, all exceeding the national average in each category.

Fig 6: Average % of Residents Who Worry That a Particular Crime May Happen To Them



Source: https://www.safewise.com/blog/safest-cities-california/ as of 04/19/2022

Another factor which makes California worth looking into is the age characteristics of its population. California has a median age of 37.3 years, which is below the national average (38.7 years), and is the 8th youngest among all U.S. states, the District of Columbia and territories[5]. Additionally, as per the United States Census Bureau, the median age of Californian males is 35.4 years. As shown in Figure 4, males within the 25-39 category make up the highest proportion of people arrested for both violent and property crime.

This paper looks at the 2012-2017 time period for the county-level analysis. The reasons for choosing this time period are multifold. Firstly, data constraints were an issue and the authors were only able to find thorough and exhaustive data for this period. Secondly, as shown in Figure 5, after 2012 California's labor market seemed to improve relative to other states as its unemployment rankings started to fall. This provides an opportunity to study whether this had any impact on crime rates. The year 2017 was added as it marked the regime change from the Obama Administration to the Trump one.

Los Angeles county is the most populated county in California. Figure 7 below analyzes the crime trend in this state within the 2012-2017 timeframe. Property crime had an average of 24 cases per 1000 residents and fluctuated between 25-22 cases. Violent crime on the other hand had an average of 5 cases per 1000 residents and generally remained constant around this value. This value is around 5 times less the average value of property crime cases (per 1000 people). It is worth comparing this to Figure 6 which showed that a higher percentage of California's residents are concerned about violent crime than property crime.

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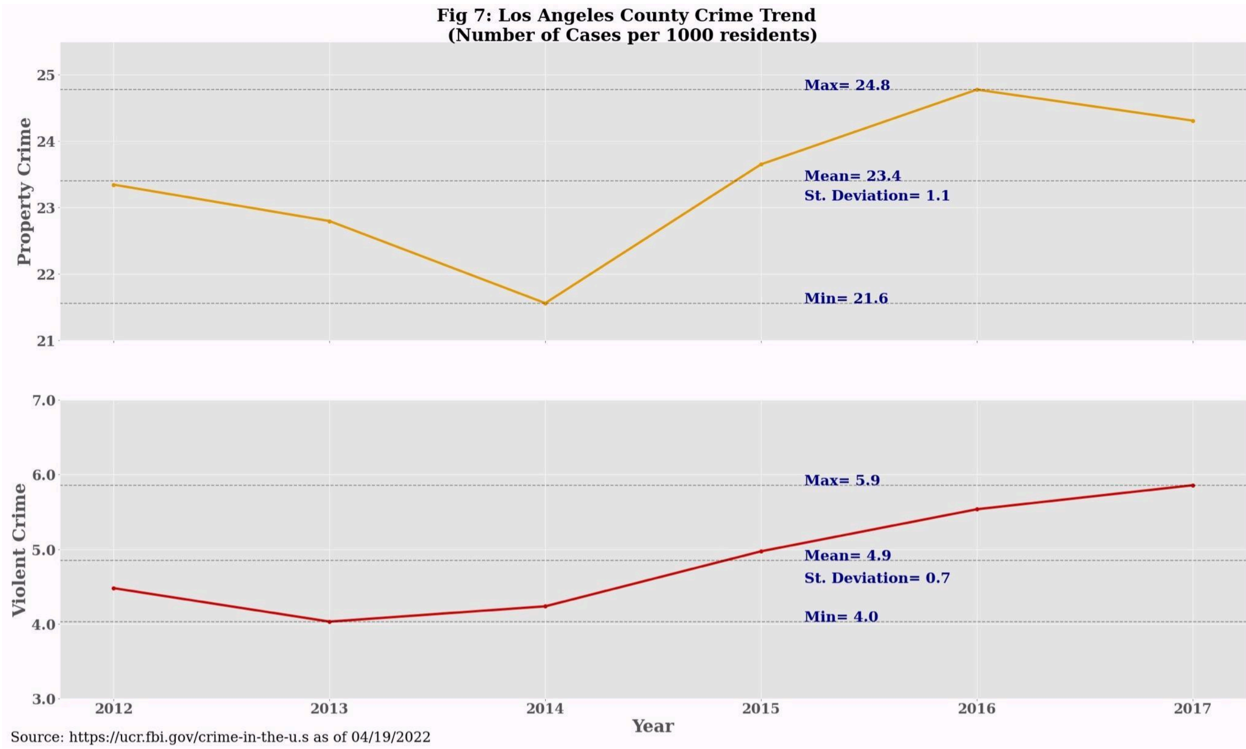
[5] Person, Johnson, H., McGhee, E., & Mejia, M. C. (2022, March 28). *California's population*. Public Policy Institute of California. Retrieved May 8, 2022, from https://www.ppic.org/publication/californias-population/



The county level analysis mainly uses the same variables as the state level analysis. However, since counties are much smaller in scale than states, therefore some changes had to be made. These changes are discussed below:

- The crime variables are now expressed as per 1000 people instead of 100,000 people.
- Previously, for the state level analysis the demographic variables (% Males 15-24 and % Males 25-39 were based on 1-year estimates. However, for counties data on such a granular basis was not available as so this analysis uses data based on 5-year estimates instead.

Additionally for the county level analysis, data from some variables was sourced from California specific data sources. These are discussed below;

- The unemployment rate for each county was sourced from the Employment Development Department of the State of California.
- For population density, population data was sourced from the California State Association of Counties. Land area is now in terms of square miles instead of square km.

The poverty rate variable is no more included. Instead we look at a new variable which is the % of population below the poverty level. The county analysis also includes two other new variables. All of the additional regressors are discussed below:

High school graduation rate: Multiple studies have identified quality education as a potentially effective way to reduce the incidence of crime. While it is true that education opens up the possibility of many alternative career paths, however, it would be unreasonable to assume that educated individuals do not commit crimes. To attempt to find an answer to this dilemma, this paper aims to assess the impact of the high school graduation rate on unemployment in the context of property and violent crime. Data for the high school graduation rate has been sourced from the Population Reference Bureau. The Alpine County was entirely dropped from the analysis due to missing data for the high-school graduation rate.

Income inequality: Income inequality is widely prevalent in California. According to the World Population Review, California with the Gini Coefficient of around 0.49 has the fourth widest income gap. Subsequently, this paper includes income inequality as a control variable in order to empirically test its significance. Data for income inequality has been sourced from the Federal Reserve Bank of St. Louis. The income inequality variable, used in this paper, is derived by calculating the ratio of the mean income for the highest quartile (top 20 percent) of earners divided by the mean income of the lowest quartile (bottom 20 percent) of earners.

% of population living below poverty line: This paper includes this control variable to evaluate whether a higher percent of population living below the poverty line has any impact on crime rates. Data for this variable has been sourced from the Federal Reserve Bank of St. Louis. It should be noted that there isn't a necessarily positive correlation between income inequality and (absolute) poverty.

Descriptive statistics for the county level analysis variables are shown below:

Table 2: County Descriptive statistics

	Mean	Standard Deviation	Minimum	Maximum	Median
Violent crime rate	4.18	1.62	0.94	8.95	4.00
Property crime rate	24.29	8.26	8.14	63.22	23.00
Unemployment rate	0.09	0.04	0.03	0.28	0.00
High school graduatIon rate	0.82	0.12	0.30	1.00	1.00
Per capita income	47,285.36	17,143.52	28,563.00	122,748.00	41,581.00
Population density	707.96	2,449.28	1.82	18,647.49	108.00
% population living below poverty line	16.25	5.06	4.90	28.30	16.00
Law enforcement employees rate	1.71	0.84	0.66	5.32	1.00
Income inequality	14.29	2.68	6.82	26.77	14.00
% Males 15-24 years	7.25	1.56	3.11	11.47	7.00
% Males 25-39 years	9.99	2.43	5.05	22.66	10.00

Model

This section discusses the model specification, the technique used and the results of the regressions. Throughout the analysis the 95% confidence interval is used to determine the statistical significance of parameters. All of the regressions are linear and the goal of this analysis is to draw inferences and aid interpretation.

This paper employs the fixed effects technique to evaluate the impact of unemployment on violent and property crime. The fixed effects technique is used as it offers the advantage of making the different states and counties comparable. This technique also assists in controlling for omitted variable bias by control for time-invariant unobserved variables (e.g Average precipitation in a region).

As an additional step the Hausman test was conducted to ensure that choosing the fixed effects technique was sensible. Results for the hausman test, shown in the table below, reinforce the choice for using this technique. As seen below, in all four cases, p-values are less than 0.05, indicating that the panel fixed effect is the consistent and efficient choice.

Table 3: Hausman Test Results[6]

Region	Variables	Hausman Results	Fixed Effect or Random Effect
US states	Violent crime	Prob>χ2=2.2e-16	Fixed Effect
	Property crime	Prob>χ2=9.355e-11	Fixed Effect
California counties	Violent crime	Prob>χ2=0.009325	Fixed Effect
	Property crime	Prob>χ2=2.2e-16	Fixed Effect

A total of four fixed effects regressions (one for each level and type of crime combination) are used. The general equation for these regressions is shown below:

$$Crime_{it} = \alpha_i + \lambda_t + \beta_1 Unemployment_{it} + \beta_2 X_{it} + \varepsilon_{it}$$

where i and t are indices for the US states/counties and year respectively. $Crime_{it}$ represents violent/property crime, α_i captures the cross sectional fixed effect, λ_t captures a year fixed effect, β_1 is the semi-elasticity of the crime rate to the unemployment rate and X_{it} indicates all the control variables.

[6] P values less than 0.05 indicate that panel fixed effect is consistent and therefore a better model fit.

Results

The section below discusses the results obtained after running our model. This section first discusses the state-level regressions (Tables 4 and 5) and then proceeds to the county level ones (Tables 6 and 7) . Two regressions are run for each case; one for violent crime and one for property crime. Parameters significant at the 95% confidence interval are highlighted in green. Parameters insignificant are highlighted in red. This paper evaluates model performance by using R^2 values and the results of the F-test for poolability (This is an F-test with a null hypothesis that all fixed effects are jointly 0). The values for these performance metrics are provided under each of the regression tables.

Results for the association between crime rates and unemployment are inconsistent. In three of the four regressions there is a strong negative correlation between unemployment and crime rates. This opposes the paper's initial hypothesis regarding a possible positive relationship between the two. However, the negative result is consistent with Cantor and Land (1985)'[7]s finding that unemployment rate negatively affects crime rates. The three regressions are both the ones in the state-level analysis and the violent crime regression for the county analysis.

In the fourth regression (The property crime one for county) the paper finds a strong positive relationship, affirming the paper's initial hypothesis. This is in accordance with prior literature such as Becker (1968)[8] and Lin (2008)[9].

Table 4 below shows the results for the State Level Violent Crime analysis. The results indicate that, at the 95% confidence level, unemployment rate has a statistically significant negative parameter. A 1 unit increase in the unemployment rate is associated with a fall in the state violent crime rate (cases per 100,000 residents) by around 5.5 units. The R^2 for this panel estimation is 34%. This indicates that unemployment and other control variables explain around a third of the variation in violent crime rates for states. In addition, the p-value for the F-test for poolability is around 0 which indicates that at the 95% confidence level the fixed effects are not jointly 0.

All control variables excluding the Full-time law enforcement employees rate, have statistically significant parameters. The paper finds that population density, poverty rate and per capita income have negative correlations. Thus it appears that more densely populated areas and higher per capita income could potentially contribute to lower incidence of violent crime.

However, the results also show that a higher poverty rate is associated with a fall in crime. These opposing results indicate that a state-level analysis might not be sufficiently insightful. Both the male variables have statistically significant positive parameters. Thus it appears, that as compared to other residents, males within these two categories are potentially much more likely

[7] Cantor, D., & Land, K. C. (1985). Unemployment and Crime Rates in the Post-World War II United States: A Theoretical and Empirical Analysis. *American Sociological Review*, *50*(3), 317. https://doi.org/10.2307/2095542
[8] Becker, G. S. (1968). Crime and Punishment: An Economic Approach. *Journal of Political Economy*, *76*(2), 169–217. https://doi.org/10.1086/259394
[9] Lin, M. J. (2008). Does Unemployment Increase Crime?: Evidence from U.S. Data 1974–2000. *Journal of Human Resources*, *43*(2), 413–436. https://doi.org/10.1353/jhr.2008.0022 . In her work, she identified that a 1 percentage point increase in the local police rate would reduce the unemployment rate by 5.1%.

to commit acts of violent crime. An important point to consider is that some categories of Violent Crime such as rape are almost always committed by males anyway.

Table 4: State Level Violent Crime---Panel OLS Estimation Summary

	Parameter	Std.Err.	T-stat	P-value
Unemployment rate	-5.5559	1.1480	-4.8397	0.0000
Population density	-0.3240	0.0702	-4.6136	0.0000
Poverty rate	-5.7873	1.3233	-4.3735	0.0000
Full-time law enforcement employees rate	-2688.5	3577.0	-0.7516	0.4526
Per capita income	-0.0033	0.0005	-6.0152	0.0000
% Males 15-24 years	79.420	13.355	5.9468	0.0000
% Males 25-39 years	48.693	9.1390	5.3280	0.0000

R-squared: 0.3394
F-test for Poolability: 125.06
P-value: 0.0000
Distribution: F(51,513)

Table 5 below shows the results for the State Level Property Crime analysis. The results indicate that, at the 95% confidence level, unemployment rate has a statistically significant negative parameter. A 1 unit increase in the unemployment rate is associated with a fall in the state property crime rate (cases per 100,000 residents) by around 16.9 units. The R^2 for this regression has a high value of 52%. This indicates that unemployment and other control variables explain more than half the variation in property crime rates for states. The p-value for the F-test is again zero, thus rejecting the null of fixed effects being jointly 0.

Again, all control variables excluding the Full-time law enforcement employees rate, have statistically significant parameters. The paper finds that, similar to the violent crime analysis, the poverty rate and per capita income variables have negative correlations. However, the population density variable now has a significant and positive parameter. Thus it appears that more densely populated areas could be potentially associated with higher property crime incidents. Again, both the male variables have statistically significant positive effects.

Table 5: State Level Property Crime---Panel OLS Estimation Summary

	Parameter	Std.Err.	T-stat	P-value
Unemployment rate	-16.867	6.78	-2.4879	0.0132

Population density	1.1636	0.41	2.8057	0.0052
Poverty rate	-20.463	7.82	-2.6185	0.0091
Employee rate	2327.0	21130.00	0.1102	0.9123
Per capita income	-0.06	0.00	-18.319	0.0000
% Males 15-24 years	398.21	78.87	5.0487	0.0000
% Males 25-39 years	311.57	53.974	5,.7725	0.0000

R-squared: 0.5221
F-test for Poolability: 50.170
P-value: 0.0000
Distribution: F(51,513)

Table 6 below shows the results for the County level violent crime analysis. The results indicate that, at the 95% confidence level, unemployment rate has a statistically significant negative parameter. A 1 unit increase in the unemployment rate is associated with a fall in the county violent crime rate (cases per 1000 residents) by around 9.21 units. The R^2 for the panel estimation for this regression is just around 18%. This indicates that unemployment and other control variables very partially explain the variation in violent crime rates for counties. The p-value for the F-test, however, is still zero.

Amongst the control variables, only 3 have statistically significant parameters. The paper finds that the percent of population below the poverty line has a negative correlation with violent crime. This is similar to the state-level result for poverty rate. However, now the results fail to identify a relationship between per capita income and violent crime.

Contrary to the results in the state-level analysis the full-time law enforcement employee rate is now indicated to have a significant positive correlation. Amongst the male variables, the 15-24 years cohort indicates significance (with a positive value) while the other one doesn't. Additionally, the results do not currently indicate a relationship with population density (again contrary to the state-level analysis), high school graduation rate and income inequality.

Table 6: County Level Violent Crime---Panel OLS Estimation Summary

	Parameter	Std.Err.	T-stat	P-value
Unemployment rate	-9.2065	2.8153	-3.2702	0.0012
High school graduate rate	0.9228	0.9952	0.9272	0.3546
Per capita income	-4.66E-06	1.52E-05	-0.3064	0.7595

Percent of population living below poverty line	-0.1001	0.0495	-2.0218	0.0442
Full-time law enforcement employees rate	1.3116	0.26	5.044	0.0000
Income inequality	-0.0118	0.0527	-0.2243	0.8227
% Males 15-24 years	0.6211	0.1923	3.2297	0.0014
% Males 25-39 years	-0.0788	0.1262	-0.6243	0.5329
Population density	-0.0004	0.0009	-0.4173	0.6768

R-squared: 0.1757
F-test for Poolability: 18.595
P-value: 0.0000
Distribution: F(56,276)

Table 7 below shows the results for the County level property crime analysis. The results indicate that, at the 95% confidence level, unemployment rate has a statistically significant positive parameter. This is the only result which affirms our hypothesis of a positive relationship between unemployment and crime rate. A 1 unit increase in the unemployment rate is associated with a rise in the county property crime rate (cases per 1000 residents) by around 60.2 units. The R^2 for this panel estimation is now higher and is around 28%. This indicates that for counties, unemployment and other control variables explain a higher variation in property crime rates. The p-value for the F-test, is again zero.

Again, only 3 control variables have statistically significant parameters. The paper finds that income inequality and population density have positive relationships with property crime. In other words, a widening economic gap and more densely populated areas are associated with a higher incidence of property crime.

For the male variables, the 25-39 years cohort indicates significance while the other one doesn't. Two points to note here are;

1. In the county violent crime regression it was the younger cohort which was significant whereas the opposite is seen in this regression.
2. Now, contrary to all three prior regressions, the significant male variable has a negative parameter. In other words a lower percentage of males, within the 25-39 age group, is associated with the incidence of crime.

The results do not currently indicate a relationship with all remaining control variables.

Table 7: County Level Property Crime---Panel OLS Estimation Summary

	Parameter	Std.Err.	T-stat	P-value
Unemployment rate	60.242	10.757	5.6001	0.0000
High school graduate rate	-2.4619	3.8028	-0.6474	0.5179
Per capita income	1.94E-05	5.81E-05	0.3346	0.7382
Percent of population living below poverty line	-0.0673	0.1892	-0.3556	0.7224
Full-time law enforcement employees rate	1.6725	0.9936	1.6833	0.0935
Income inequality	0.4348	0.2014	2.1589	0.0317
% Males 15-24 years	0.6707	0.7348	0.9128	0.3622
% Males 25-39 years	-1.212	0.4823	-2.5132	0.0125
Population density	0.013	0.0035	3.7029	0.0003

R-squared: 0.2830
F-test for Poolability: 21.057
P-value: 0.0000
Distribution: F(56,276)

Limitations and Recommendations for Further Review

This analysis doesn't fall short of limitations. Due to data inaccessibility, this paper does not include other relevant factors such as average ground temperature and annual precipitation. Environmental stressors and extreme weather events could potentially impact crime and are also expected to be correlated with some of the independent variables such as per capita income. Therefore our analysis is susceptible to omitted variable bias. Since these factors are not time invariant due to climate change, therefore, the fixed effects technique also fails to control them.

Another limitation of the model specification is that this paper uses a linear model to make the analysis more straightforward. The impact of different rates on each other is more likely to be non-linear and perhaps could better be explained through quadratic or higher power variables. As is a frequent problem with fixed effects techniques, the regressions in this paper suffer from low statistical power, as is evident from the relatively low R values.

In terms of the sample, for states this paper only considers the period of 2006 to 2016 due to the lack of crime data from the FBI. For counties, the impact of unemployment on crime is only accessed for a 5-year span, again due to data unavailability. The Alpine county had to be removed from the analysis as data wasn't available for it. In addition, this paper only looks at the county of California.

While this paper identifies certain relationships between crime and unemployment, this paper fails to provide any sociological or theoretical context for them as this wasn't the area of expertise of any of the authors. This paper also fails to provide a consistent direction of relationship which points towards the need for further in-depth review

These limitations leave significant room for improvement. Further review should focus on including more variables, using non-linear models, looking at longer time frames and including more counties. It would also be prudent to look at the theoretical relationship between unemployment and crime by collaborating with sociologists or other non-economist experts to draw conclusions. Counties for other states should also be reviewed.